

Simon Syga

Evolutionary Dynamics · Phenotypic Plasticity · Spatial and Multiscale Modeling · Mathematical Oncology

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Academic Appointments

Postdoctoral Researcher , Center for Interdisciplinary Digital Sciences, TU Dresden	02/2024–present
Mathematical modeling of cancer therapy response; scientific lead for modeling in the ERC Advanced Grant <i>subLETHAL</i> .	
Doctoral Researcher , Center for Interdisciplinary Digital Sciences, TU Dresden	03/2018–02/2024
Agent-based and continuum modeling of tumor evolution, phenotypic plasticity and cancer cell invasion; epidemic modeling and transmission-network analysis during the COVID-19 pandemic. Developed the <code>bio1gca</code> Python package for cellular-automaton simulations.	
Guest Researcher , Centre de Recerca Matemàtica, Universitat Autònoma de Barcelona	10/2023–12/2023
Cellular-automaton modeling of angiogenesis with phenotypic plasticity.	
Research Assistant , Center for Information Services and HPC, TU Dresden	12/2016–02/2018
Cellular-automaton modeling of collective cancer-cell migration.	
Research Assistant , Max Planck Institute for the Physics of Complex Systems	03/2016–06/2016
Agent-based modeling of biaxial nematic interactions in complex geometries.	
Undergraduate Assistant , Max Planck Institute for the Physics of Complex Systems	04/2014–07/2014
Agent-based modeling of uniaxial nematic interactions of hepatocytes.	

Education

PhD in Mathematics (summa cum laude) , TU Dresden	03/2018–02/2024
Dissertation: <i>Impacts of genetic and phenotypic heterogeneity on tumor evolution</i> . Supervisors: Prof. A. Voigt and Prof. A. Deutsch.	
Master of Science in Physics , TU Dresden	10/2014–01/2018
Thesis supervisors: S. Grill and A. Deutsch. Grade: 1.0 (ECTS grade A, top 5%).	
Semester Abroad , Lund University, Sweden	01/2015–07/2015
Bachelor of Science in Physics , TU Dresden	10/2011–11/2014
Thesis supervisors: F. Jülicher and S. Grill. Grade: 1.3 (ECTS grade A, top 5%).	

Independent Funding and Fellowships

Postdoc Starter Kit , Graduate Academy, TU Dresden	2025–2026
€10,000 for research visits, collaboration and conference activity.	
State Innovation Scholarship , European Social Fund and Free State of Saxony	2018–2021
Doctoral scholarship.	

Research Leadership in Collaborative Grants

ERC Advanced Grant <i>subLETHAL</i> — scientific lead for modeling	2022–2027
Collaborative project <i>From understanding to rational design of next-generation cancer therapies</i> . PI: P. Friedl, Radboud University Nijmegen; total award: €2.5 million.	
FOSTER.AI student research project <i>Morpheus.AI</i> — project supervisor	2026–2027
€10,000 funding for an AI-assisted mathematical-model development workflow.	
Worldwide Cancer Research — scientific lead for modeling	2023–2025
Collaborative project <i>Dissecting and overcoming collective therapy resistance</i> . PI: P. Friedl, Radboud University Nijmegen; total award: £200,000.	

Honors, Awards, and Competitive Academic Selection

W1 Junior Professorship in Biomathematics, University of Greifswald Ranked second on the formal appointment list.	2026
Lindau Nobel Laureate Meeting Competitively selected participant following an international multi-stage process.	2026
Integrated Mathematical Oncology Workshop, Moffitt Cancer Center Member of the runner-up team awarded a \$50,000 pilot grant for a project on personalized therapy in ovarian cancer.	2025
German Thesis Award National doctoral thesis award, shortlisted (top 12 of 254 in Natural and Technical Sciences), Körber Foundation.	2025
Reinhart-Heinrich Doctoral Thesis Award International doctoral thesis award, European Society for Mathematical and Theoretical Biology.	2024
MTZ Award for Systems Medicine National doctoral thesis award, MTZ Foundation; €3,300 .	2024
Poster Prize Society for Mathematical Biology Annual Meeting.	2019
Travel Grants (8) Moffitt Cancer Center, Erasmus+, European Society for Mathematical and Theoretical Biology, Graduate Academy of TUD, Association of Friends and Sponsors of TUD.	2018–2026

Selected Invited Talks and Award Lectures

World Congress on Computational Mechanics, Munich — minisymposium talk	2026
European Conference on Mathematical and Theoretical Biology, Graz — Reinhart-Heinrich Award lecture	2026
Max Planck Institute for Evolutionary Biology, Plön — workshop talk	2025
Mathematical Oncology Laboratory, Ciudad Real — seminar talk	2025
Moffitt Cancer Center, Tampa — seminar talk	2024
Conference on Systems Biology of Mammalian Cells, Leipzig — MTZ Award lecture	2024
Max Planck Institute for Evolutionary Biology, Plön — seminar talk	2023
Additional invited and contributed presentations at international conferences and workshops including <i>European Conference on Mathematical and Theoretical Biology</i> , <i>Society for Mathematical Biology</i> , and <i>Deutsche Physikalische Gesellschaft</i> .	

Supervision and Mentoring

Doctoral supervision

P. Chandratre , <i>Supporting Physical and Computational Biology with AI-Powered Multi-Agent Model Generation</i> Computer Science, TU Dresden; scientific supervisor; degree supervisors: Prof. M. Weigert and Prof. W. Nagel.	2026–present
E. Weinholtz , <i>Graph-Based Analysis and Modeling of Cancer Cell Interactions and Therapy Resistance</i> Mathematics, TU Dresden; scientific supervisor; degree supervisors: Prof. A. Voigt and Prof. A. Deutsch.	2023–present

Master's and bachelor's supervision

Co-supervision of seven MSc theses and seven student research projects or internships since 2019; several projects contributed to peer-reviewed publications. Selected projects:	
A. Whiteside , <i>Evolution of Phenotypic Plasticity in Glioblastoma</i> One-year pre-doctoral research project, TU Dresden.	2025–2026
P. Chandratre , <i>Evaluation and Optimization of an LLM-Based Agentic Workflow for Generating Multicellular Models</i> MSc thesis, TU Dresden.	2025–2026
M. Nussbaumer , <i>Mathematical Modeling of Early-Stage Tumor Dynamics</i> MSc thesis, TU Dresden.	2024–2025
T. Paulat , <i>Invasion-Associated Radiotherapy Resistance of Cancer</i> MSc thesis, TU Dresden.	2021–2022